MASooda — P3 Boids & Corridor Formulation (Week 1, final)

1. Conventions

- Boids acts in the horizontal plane. Altitude is a discrete layer owned by P1/P4, so desired $v_z = 0$.
- $p_i, v_i \in \mathbb{R}^2$ are the (x, y) position and velocity of agent *i*. N_i is its neighbor set from P1's communication graph (40 m radius / k = 6 NN). $d_{ij} = \|p_i - p_j\|$.
- An agent's **corridor** is the corridor whose centerline is nearest to it. $S_i \subset N_i$ is the set of neighbors in the same corridor as *i*.
- Output is a desired velocity. P1 applies the v_{max} and a_{max} clamps. P3 also saturates to v_{max}.
- No parameter is hard-coded. Everything lives in the `corridors` and `local_behavior` sections of `base_config.yaml`.

2. Corridor geometry

Three parallel corridors along +x, centerlines at $y_k = k \cdot s$.

Parameter	Value
Number / width / capacity	3 / 20 m / 15 (base_config)
Centerline spacing s	50 m (30 m empty strip, well above d_min = 8 m)
Length L	1500 m
Entry / merge / exit zones	x ∈ [0,100] / [700,800] / [1400,1500]

Changing corridor (decided):

- Allowed only inside the merge zone, which is also where P2's ADMM is triggered.
- P4's overseer decides the target corridor (`corridor_map`). P3 only executes it. An agent never changes lane on its own.
- One corridor step at a time (0↔1, 1↔2).
- Executed by switching the agent's target centerline, so it steers sideways while moving forward.
- Feasibility: 50 m sideways over 100 m at ~12 m/s forward needs ~6 m/s lateral, about 13 m/s total, under 15 m/s. If it is too tight, lengthen the merge zone to 150 m.

3. Behavior terms

3.1 Separation (all neighbors, including other corridors, which keeps lane changes safe)

```
v_sep = Σ_{j∈N_i, d_ij < r_sep} (p_i - p_j) / d_ij²
r_sep = 20 m
```

Optional extra: scale each term by `1 + β·max(0, prio_i - prio_j)` so lower-priority agents yield more.

3.2 Alignment (same corridor only)

```
v_ali = (1/|S_i|) Σ_{j∈S_i} v_j - v_i
```

3.3 Cohesion (weak, along-corridor only)

```
c_i   = (1/|S_i|) Σ_{j∈S_i} p_j - p_i
v_coh = (c_i · ĥ) ĥ ,   ĥ = (1, 0)
```

Switched off (`v_coh = 0`) if any same-corridor neighbor is closer than `r_sep`.

3.4 Corridor following

```
q_i    = (min(x_i + ℓ, L), y_target)
ℓ = 40 m look-ahead
v_goal = v_cruise · (q_i - p_i)/‖q_i - p_i‖ + [0, -k_lat·(y_i - y_target)]ᵀ
```

If no corridor is known, `q_i` is the destination.

3.5 Platooning: safe gap, slow-down only (decided) Leader = nearest same-corridor neighbor ahead (`x_j > x_i`) within `r_plat`.

```
gap    = x_j - x_i
gap*   = d_0 + h·‖v_i‖                          (≈ 24 m at 12 m/s)
v_plat = min(0, k_s (gap - gap*) + k_v (v_j,x - v_i,x)) · ĥ
```

The agent can only be slowed, never sped up to close a gap. `k_s` has units 1/s, so `v_plat` is a velocity (m/s) and no extra `Δt` factor is needed. (My earlier draft had one by mistake.)

3.6 Combination

```
v_des = w_g v_goal + w_sep v_sep + w_ali v_ali + w_coh v_coh +
v_plat
v_des ← v_des · min(1, v_max/‖v_des‖)
output = [v_des,x, v_des,y, 0]
```

Priority when terms conflict: separation > corridor following > platooning > alignment > cohesion.

4. Parameters (starting values, tuned in Week 2)

Parameter	Value
`v_cruise`	12 m/s
`r_sep`	20 m
`w_g, w_sep, w_ali, w_coh`	1.0, 2.5, 0.4, 0.1
`k_lat`	0.5 s^{-1}
ℓ (look-ahead)	40 m
`d_0` / `h`	12 m / 1.0 s
`k_s` / `k_v`	0.3 s^{-1} / 0.5
`r_plat`	40 m
β (optional priority yield)	0.5

5. Failure behavior

Condition	Behavior
Directive is `None`	Use the last-known corridor (cached per agent)
No neighbors	Goal seeking only
No corridor ever assigned	Head toward `destination`
NaN or zero-norm vector	Treat as zero contribution
Under attack (Week 4)	Use P5's W-MSR-filtered neighbor estimates instead of raw broadcasts

6. Open points for the Week-1 sync

1. **Capacity:** 3 × 15 = 45 slots but 60 agents. P4/P1 must decide on entry throttling or per-layer capacity.
2. **Altitude:** corridors are 2D lanes and layers only add vertical separation. Confirm with P1/P4.
3. **Config:** P1 adds the `corridors` and `local_behavior` sections above to `base_config.yaml`.

Glossary and definitions

(Place this before "1. Conventions".)

Terms

Term	Definition
Agent	One drone (emergency, eVTOL, or delivery).
Boids	A flocking model in which each agent steers using simple local rules: separation, alignment and cohesion.
Separation	Boids rule that pushes an agent away from neighbours that are too close.
Alignment	Boids rule that makes an agent match the average velocity of its neighbours.
Cohesion	Boids rule that pulls an agent toward the centre of its neighbours.
Corridor	A predefined 20 m wide lane in the horizontal plane. There are three, running along +x.
Corridor assignment	The corridor the overseer has told an agent to use (`corridor_map`).
Corridor following	Steering an agent toward the centerline of its target corridor while moving forward.
Entry / merge / exit zone	Stretches of the corridor by x-position. Agents enter in the entry zone, may change corridor only in the merge zone, and leave in the exit zone.
Lane change	Moving from one corridor to an adjacent one. It is allowed only in the merge zone and only as the overseer directs.
Platoon	A group of agents following each other along a corridor.
Time headway	The time gap between an agent and the one ahead. The desired distance grows with speed.
Leader	The nearest same-corridor neighbour ahead of an agent (larger x).
Neighbour set	The agents within communication range of agent *i*, from P1's communication graph (40 m radius or k = 6).
Same-corridor neighbour	A neighbour whose nearest corridor centerline is the same as agent *i*'s.

Priority	An integer where lower means more important: emergency 0, eVTOL 1, delivery 2.
Overseer directive	P4's output. It contains the corridor map and the priority table.
Cached directive	The last directive received, reused if the overseer link is down.
Desired velocity	The velocity P3 asks for. P1 then applies its own limits and integrates it.
Altitude layer	A discrete height level, owned by P1/P4. P3 does not control altitude, so desired `vz = 0`.
ADMM	P2's method for resolving conflicts in the merge zone.
STL	Signal Temporal Logic, used by P6 to check the minimum-separation rule.

Symbols

Symbol	Meaning	Unit
`i`, `j`	Agent indices	–
`p_i`	Horizontal position (x, y) of agent *i*	m
`v_i`	Horizontal velocity of agent *i*	m/s
`x_i`, `y_i`	Components of `p_i`	m
`v_j,x`	x-component of neighbour *j*'s velocity	m/s
`d_ij`	Distance between agents *i* and *j*, $\|\| p_i - p_j \|\|$	m
`N_i`	Neighbour set of agent *i*	–
`S_i`	Same-corridor neighbours of *i*, a subset of `N_i`	–
`prio_i`	Priority number of agent *i*	–
$\hat{h}$	Unit vector along the corridor, (1, 0)	–
`k`	Corridor index (0, 1, 2)	–
`s`	Spacing between corridor centerlines (50 m)	m
`y_k`	y-position of corridor *k*'s centerline, $k \cdot s$	m
`y_target`	Centerline y of the agent's target corridor	m

`L`	Corridor length (1500 m)	m
`ℓ`	Look-ahead distance (40 m)	m
`q_i`	Look-ahead target point on the target centerline	m
`d_min`	Minimum allowed separation checked by STL (8 m)	m
`r_sep`	Separation radius (20 m)	m
`r_plat`	Range within which a leader counts for platooning (40 m)	m
`v_max`	Maximum speed (15 m/s)	m/s
`a_max`	Maximum acceleration (4 m/s²)	m/s²
`v_cruise`	Preferred cruising speed (12 m/s)	m/s
`gap`	Distance to the leader along the corridor, `x_j - x_i`	m
`gap*`	Desired gap, `d_0 + h·‖v_i‖`	m
`d_0`	Standstill gap, the desired gap at zero speed (12 m)	m
`h`	Time headway (1.0 s)	s
`k_s`	Gap-error gain	1/s
`k_v`	Speed-error gain	–
`k_lat`	Lateral correction gain toward the centerline	1/s
`β`	Optional priority-yield gain for separation	–
`c_i`	Vector from agent *i* to its same-corridor neighbours' centre	m
`v_goal`	Corridor-following velocity term	m/s
`v_sep`, `v_ali`, `v_coh`, `v_plat`	Separation, alignment, cohesion and platooning velocity terms	m/s
`w_g`, `w_sep`, `w_ali`, `w_coh`	Weights on the goal, separation, alignment and cohesion terms	–
`v_des`	Final desired velocity of the agent	m/s

One note on the units: `v_sep` as written, `Σ (p_i − p_j) / d_ij²`, has units of 1/m, not m/s. It is a relative strength, so `w_sep` implicitly absorbs the conversion. That's fine for tuning in Week 2, but if you want the units to be clean, `w_sep` could carry m²/s instead of being dimensionless. Tell me if you'd like the table updated that way.